Abdullah Al Zahid

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SUMMARY

Transportation engineer with a master's degree and four years of GIS and data work across municipal transportation planning at the City of Toronto and GO rail service planning at Metrolinx. Builds the spatial datasets a division depends on and turns scattered sources into one validated source people report from.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Build and maintain the road and right-of-way datasets the division depends on, using GIS and Python, including a ROW-width dataset covering all city roads, built with QGIS, MapBox and JavaScript.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.
- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Run assigned studies from scoping through to recommendations, writing the briefing notes that come out of them and presenting to planners, engineers and managers in other divisions.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Ran OpenTrack simulations of GO rail services to establish what proposed operating and infrastructure changes would do to run times, reliability and recovery before they were taken forward.
- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.
- Worked with schedulers, operations staff and engineers on service design decisions and presented the evidence behind them.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Spatial analysis of large transportation datasets in GIS, turning data into recommendations for safer wildfire evacuation plans in Alberta communities.

SKILLS

ArcGIS Pro, QGIS and MapBox. Python (Pandas, NumPy), SQL and JavaScript for spatial data processing and automation. Data integration, validation and dashboards. OpenTrack, VISSIM, Synchro and MATSim. AutoCAD and Civil 3D. Advanced Excel, including large multi-source workbooks. Briefing notes and presentations for senior staff.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022. CGPA 4.0/4.0; fully funded by the university; graduate course in GIS.

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018. CGPA 3.69/4.0; University Merit Scholarship, University Stipend Scholarship, Technical Scholarship, Dean's List.

BSc, Computer Science — University of British Columbia, in progress (2025–2027)